# Project Report

on

## Weather Forecast Web Application

in partial fulfilment for the award of the degree of

**BACHELOR OF ENGINEERING**

IN

**COMPUTER SCIENCE**


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## 1.Problem Statement

The objective of this project is to design and implement an efficient **Weather Forecast Web Application** capable of providing real-time weather information for any city or the user's current location. The system enables users to search weather conditions dynamically, access location-based forecasts, and view important climate parameters in a simple and user-friendly interface.

A major challenge in weather applications is fetching accurate real-time data from external APIs and presenting it instantly with proper error handling. Therefore, the system must manage API requests efficiently, handle invalid city names, support browser geolocation permissions, and display weather results in an organized responsive design.

## Objectives of the Project

- To develop a system that fetches live weather data using the **OpenWeather API.**
- To implement **current location weather detection** using browser geolocation.
- To allow users to search weather details for any city dynamically
- To display weather parameters such as temperature, humidity, wind speed, and cloudiness.
- To design a responsive and attractive user interface using **HTML, CSS, and JavaScript**

# 1. ABSTRACT

This project focuses on the design and implementation of a **Weather Forecast Web Application** that efficiently provides real-time weather updates for users. The system is developed using modern web technologies and integrates with the **OpenWeather API** to fetch accurate weather information for any city or the user's current location.

The frontend is implemented using **HTML, CSS, and JavaScript**, providing an interactive single-page interface with two main features: **Your Weather** (current location weather) and **Search Weather** (city-based weather search). The application uses browser geolocation services to detect the user's coordinates and display local weather instantly.

The system retrieves live weather data such as:

- Temperature
- Weather condition description
- Humidity
- Cloud Percentage

Overall, this project also includes loading animations, responsive design, session storage for saved coordinates, and error handling for invalid city names or failed API requests.

## 2. Flow Of the App

### 2.1 Current Location Weather Fetch

The system uses the browser geolocation feature to detect the user's live coordinates and fetch weather data.

1. Start the process.
2. Check whether browser supports geolocation.
3. If supported, request latitude and longitude from user permission.
4. If permission granted, store coordinates in session storage.
5. Send API request to OpenWeather using latitude and longitude.
6. Receive JSON weather response.
7. Extract city name, temperature, humidity, wind speed, clouds, weather icon.
8. Display results on screen.
9. If permission denied, show grant access message.
10. End process.

### 2.2 City Search Weather

1. Start the process.
2. User enters city name in search box.
3. Validate input field is not empty.
4. Send API request using city name.
5. Wait for response.
6. If city exists, parse response data.
7. Display weather details on screen.
8. If city is invalid, show error image/message.
9. End process.

## 2.3 Weather Data Rendering

1. Receive API response data.
2. Extract city name and country code.
3. Generate country flag URL.
4. Extract weather description and icon code.
5. Extract temperature value.
6. Extract humidity percentage.
7. Extract wind speed.
8. Extract cloud percentage.
9. Update DOM elements dynamically.
10. Display weather card to user.

# 3. FLOWCHART

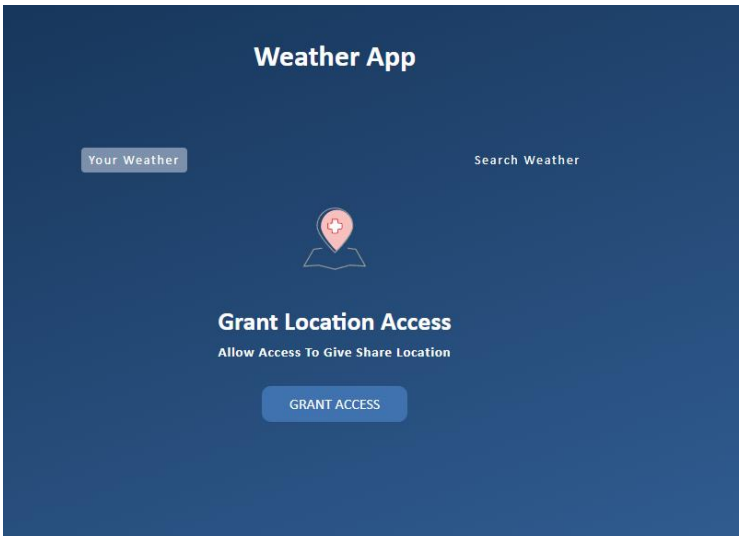
3.1 Current Location Weather Flowchart

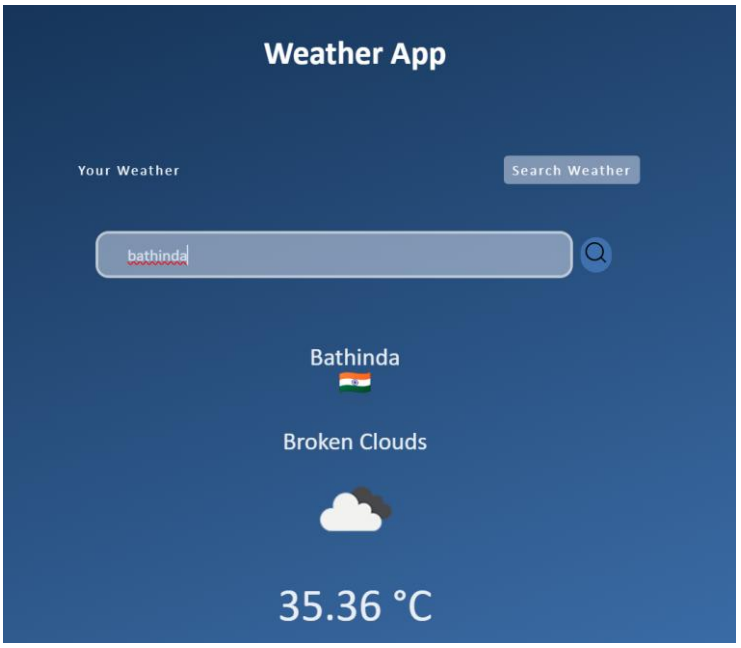
```
            START
              |
              v
   Check Geolocation Support
              |
      +------+------+
      |             |
     Yes           No
      |             |
      v             v
Ask User Permission Show Error
      |
      v
Get Latitude & Longitude
      |
      v
Send API Request
      |
      v
Receive Response
      |
      v
Display Weather Data
      |
      v
             END
```

## 3.2 Search Weather Flowchart

```
            START
              |
              v
    Enter City Name
              |
              v
     Validate Input
              |
      +------+------+
      |             |
   Empty         Valid
      |             |
      v             v
 Stop Process   Send API Request
                     |
                     v
                Receive Response
                     |
               +-------+------+
               |              |
            Success        Invalid
               |              |
               v              v
     Display Weather     Show Error
               |
               v
              END
```

## 3.3 ScreenShots



## 4.CONCLUSION

The **Weather Forecast Web Application** successfully demonstrates the practical implementation of modern web development concepts through real-time API integration and responsive user interface design. By combining **HTML, CSS, and JavaScript** with the **OpenWeather API**, the system is capable of providing accurate weather information for both the user's current location and searched cities instantly.

The project highlights the importance of integrating external APIs into web applications to solve real-world problems efficiently. Features such as geolocation access, city-based search, loading indicators, dynamic weather icons, and error handling ensure a smooth and user-friendly experience.

The modular design of the application, with separate sections for UI structure, styling, and JavaScript functionality, improves maintainability and allows future upgrades easily. The use of session storage for saving user coordinates further enhances usability by reducing repeated permission requests.

This project also demonstrates key frontend development skills such as asynchronous programming using Fetch API, DOM manipulation, event handling, responsive design, and API data rendering. These are highly valuable skills in full stack development.

The Weather App provides a strong foundation in frontend and API-based project development, and demonstrates how real-time data services can be transformed into a practical, interactive, and visually appealing web application.

## Future Scope

- Developing a full graphical dashboard with charts for **temperature, humidity, wind speed, and rainfall trends** using JavaScript chart libraries.
- Adding support for **multiple saved favorite cities** for quick weather access.
- Integrating a backend server (**Node.js / Spring Boot / Python Flask**) to store user preferences and search history.
- Implementing **AI-based weather recommendations**, such as clothing suggestions, travel advice, or outdoor activity alerts.
- Exporting weather reports in **PDF, CSV, or printable formats** for personal and business use.
- Adding **severe weather alerts** such as storms, heavy rain, heatwaves, and air quality warnings.
- Supporting **multiple languages and voice search** for better accessibility.
- Creating a dedicated **mobile app version** for Android and iOS platforms.
- Integrating maps to show **live weather conditions by region or country** visually.